De La Salle Holy Cross College

NOVEMBER EXAMINATION

2024

Grade 10

SPORTS AND EXERCISE SCIENCE

Examiner:	Mr S Moore	**Time:**	2 hours
Moderator:	Mrs D Webb	**Marks:**	160

Name:

PLEASE READ THROUGH THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 29 pages and a separate Source Material booklet of 4 pages. Please check that your question paper is complete.
2. **Answer all the questions on the question paper** and hand it in at the end of the examination.
3. Use the total marks awarded for each question as an indication of the detail required.
4. It is in your own interest to write legibly and to present your work neatly.
5. FOUR blank pages (pages 26 – 29) are included at the end of the question paper. If you run out of space for a question, use these pages. Clearly indicate the number of your answer should you use this extra space.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	
Mark									
Total	**10**	**20**	**30**	**20**	**22**	**20**	**18**	**20**	**160**

QUESTION 1

Write the correct answer in column B to match the correct description in column A.

COLUMN A	COLUMN B
1.1 This type of muscle is controlled voluntarily.	Skeletal/ striated muscle
1.2 This muscle is named after the Greek letter delta.	deltoid
1.3 This tissue connects muscle to bones.	tendons
1.4 The term *Maximus* means the muscle is...	Big/ Great
1.5 This muscle is the antagonist for knee flexion.	Quadriceps
1.6 This muscle is the antagonist for elbow flexion.	Tricep
1.7 This joint movement is when a limb moves sideways to the body …	Abduction/ Adduction
1.8 The term *Inferior* refers to…	Towards the feet
1.9 The term *Supine* means…	On back facing sky
1.10 Abduction occurs in this body plane.	Frontal

[10]

QUESTION 2

Examine Picture A and answer the questions that follow.

Picture A



www.skeletal.com

2.1 Identify one bone that the Olympic Bar is resting on.

Scapula/ Metacarpals_____(1)

2.2 List three bones in the upper limb.

Scapula, Humerus, Ulna, Radius accept feasible.

_____(3)

2.4 Name two minerals that are stored in bones.

Calcium and phosphate

_____(2)

2.5 Name the joint movement occurring at the knee in Picture A.

Knee flexion____(2)

2.6 Based on the answer in 2.5 name the muscles acting as the agonist and antagonist for that joint movement.

Agonist- hamstring

Antagonist- Quadricep

_____ (4)

2.7 If the weightlifter remains in that position for 1 minute, name the type of muscle contraction that is occurring and describe what is happening to the muscle.

Isometric contraction

Muscle is under tension without changing length

_____(3)

2.8 List 5 physiological adaptations that will occur to the **Musculo-Skeletal System only** as a result of doing the exercise in Picture A for a prolonged period of time.

Accept any 5

Increased myoglobin

Increased mitochondria

Muscle Hypertrophy/ growth

Increased anaerobic capacity/ enzymes

Increased Glucose Oxidation

Increased ATP PC stores

_____

_____(5)

[20]

QUESTION 3

Examine Source A and answer the questions that follow.

Source A

How Energy Systems Training Can Help You Be Better at Your Sport

When athletes and coaches think about conditioning, they often picture endless bouts of Suicide Sprints, Stair Runs and other running drills performed with reckless abandon. Showing up to practice on day one and performing sprints for hours is almost like a rite of passage.

This way of thinking will only get you so far. Better to understand the energy systems in your body and how to train and best use them for your sport.

Many people still believe the aerobic system is only for marathon runners. However, you need to have an aerobic base and build upon it.

The best way to think about your energy systems is to picture a house. You have your basement/foundation, your first floor with the kitchen, bedrooms, and living room and finally your attic. Without a good foundation, the house will eventually crumble. If all you do is high-intensity exercise, you will eventually run out of gas.

Developing a foundation helps us develop more aerobic enzymes, left ventricle eccentric hypertrophy (increased stroke volume and decreased resting heart rate) and mitochondria, among other advantageous adaptations. Our aerobic system allows us to compete for along duration, recover between high-intensity work bouts, buffer our muscles for energy production and maintain power output. Without it, we would all be really great for 15-60 seconds and eventually fall apart.

https://news.umich.edu/

3.1 With reference to the house analogy in Source A, provide the names of the related energy systems.

Basement/ Foundation Aerobic/ Oxidative

First Floor Lactic Acid System

Attic ATP PC System (6)

3.2 Name the organelle from which ATP is produced.

Mitochondria (1)

3.3 Provide definitions for Aerobic and Anaerobic energy system.

Aerobic: Exercise which uses 02 to produce energy

(2)

Anaerobic: Exercise which does not use 02 to produce energy

(2)

3.4 Complete the empty blocks below to show a comprehensive view of energy systems in relation to sporting activities.

(18)

Energy System	**Fuel used**	**Duration**	**Sporting Example**

ATP/PC system	ATP/PC	0-15 SECONDS	100m Sprint
Lactic Acid/ Anaerobic Glycolosys	Glucose/ Glycogen	15 secs –3 min depending on intensity.	400m sprint
Aerobic/ Oxidative	Carbohydrates and Fats	3 min and more	Marathon

3.5 What percentage (range) of ones maximum heart rate could one expect to be in when exercising aerobically?

40-60%

(1)

[30]

QUESTION 4

Examine Source B and answer the questions that follow.

Source B

Is Pull the right term for our arms in Swimming?

I was recently asked if using the term pull for the arm motion in Swimming is the right term. Another term that was suggested for the arm motion was the word leverage. Here is why I disagree with using the term leverage for a swimmer's arm motion.

The word leverage implies the use of a lever. In most cases, we think of levers as something that increases the mechanical advantage- enabling us to use a small force to lift or move big objects, but they don't always do that.

The arm is definitely used as a lever in Swimming. In fact, it is a Class III lever. That means that the effort load (muscles moving the arm and hand through the water) and the resistance load (mainly applied to the swimmer's hand) are located on the same side of the lever's fulcrum (shoulder joint).

With the arm, the distance between the fulcrum (shoulder joint) and the effort load (insertion of the muscles) is called the effort arm. The distance is very short, perhaps an inch or two, as the muscles moving the upper arm insert into the head of the humerus bone. The distance from the fulcrum (shoulder) to the resistance load, or hand, is called the load arm. That distance is much longer- more than 30 inches in adults.

https://www.healthline.com/

4.1 Provide a definition of a lever.

Rigid bar

That rotates around a fixed point

Called a fulcrum

Or

A simple machine that provides a mechanical advantage

_____(3)

3.2 Provide the definition of an " Effort Arm".

Distance between the fulcrum and the effort.

_____(3)

3.3 Provide a definition for a "Load Arm".

Distance between the fulcrum and the load/ resistance.

_____ (3)

4.2 Identify the Effort, Load and Fulcrum in the below two pictures. Once identified, state the class of lever being used.



Effort Any muscle in the upper limb

Load Javelin

Fulcrum Any joint in the upper limb

Class Correct application of the FRE principle (4)



Effort Any muscle in the upper limb

Load Javelin

Fulcrum Any joint in the upper limb

Class Correct application of the FRE principle (4)

4.3 Match the Anatomical Plane in column A to the corresponding movement that occurs in that plane in column B. Write only the letter of your chosen description in the answer table.

COLUMN A		COLUMN B

4.3.1	Frontal Plane	**A.**	Cervical Flexion
4.3.2	Transverse Plane	**B.**	Hip Abduction
4.3.3	Saggital Plane	**C.**	Thoracic Rotation
		D.	Hip Extension
		E.	Cervical Extension
		F.	Hip External Rotation

ANSWER TABLE

Term	**Letter**
Frontal Plane	**B**
Transverse Plane	**C or F**
Saggital Plane	**A, D OR E**

(3)

QUESTION 5

Examine Source C and answer the questions that follow.

Source C

Carnivore Diet and Endurance Training
Maximizing Performance Naturally

The Carnivore Diet, characterized by an all-meat regimen, is gaining attention in the athletic community, particularly among those involved in endurance training. At its core, this diet consists predominantly of animal products, omitting traditional sources of carbohydrates such as fruits, vegetables, and grains. Advocates suggest that the high protein content integral to the diet aids in muscle repair and recovery, which could be beneficial for athletes who engage in prolonged physical activity. They argue that by prioritizing protein and fats over carbohydrates, an athlete's body can adapt to efficiently utilize fat as a primary energy source, which could potentially enhance performance during endurance sports.

While the Carnivore Diet presents a paradigm shift from conventional endurance training nutrition, its actual influence on athletic performance is an area of ongoing exploration. Athletes contemplating this diet are encouraged to consider both the potential benefits of high-protein intake for muscle repair and the implications of a limited nutrient profile. It is essential for each athlete to assess the diet's alignment with their individual health needs and performance goals, often consulting with a sports nutritionist to ensure the chosen regimen sustains their rigorous training demands and promotes their overall well-being.

https://discover.texasrealfood.com/

5.1 Provide the definition for the term ***calorie***.

The amount of energy required to heat 1l of water by 1 deg calcius.

_____(3)

5.2 Provide the correct *calorific value* for the standard amount of energy provided by each of the following:

.

Carbohydrate 4kcal/g

Fat 9kcal/g

Protein 4 kcal / g
(6)

5.3 Would the carnivore diet be classified as " High GI" or "Low GI"? Justify your answer by defining and explaining the term " GI".

GI stands for Glycaemic Index. (2)

Refers to the rate/ speed at which glucose enters the blood stream. (2)

Carnivore is Low GI. (1)

There are no carbohydrates/ glucose in meat and fat. (1)

_____

_____

_____

_____

_____(6)

5.4 When should an athlete consume a high GI meal vs a Low GI meal? Justify your answer.

High GI- during exercise to keep glucose levels sufficient for muscle energy production

Or after exercise keep glucose levels sufficient for recovery

Low GI- before exercise- to prevent glucose/ insulin spikes, which results in an glucose crash. Glucose crashes are negatively associated performance.

_____

_____(4)

5.5 Which list three important functions that fat is responsible for with regard To the nervous system.

Insulates nerves
Transports Vitamins
Builds cell membranes
Protects organs (3)

[22]

QUESTION 6

Examine Source C and answer the questions that follow.

Source C

The Science Of Lance Armstrong

I have heard a lot of comparisons between Michael Phelps and Lance Armstrong. I have to admit I don't know too much about swimming, but I do know about cycling, particularly Armstrong, here are some of the things that i know. The main reason for Armstrong's dominance in his sport is something most great endurance athletes have in common. Yet when it comes to lance's own individual physiology, he is just nearly superhuman. Here's why:

His genetics for an endurance athlete are nothing short of astounding. Do you know his VO2 Max measures around 83-85 ml/kg/min, while an average person's is around 40? For Lance, his success is attributed to a combination of great training, will, desire, and unbelievable genetic makeup.

He also has a very large heart, his over sized heart can beat over 200 times a minute and thus pump an extraordinarily large volume of blood and oxygen to his legs. His VO2 max—the maximum amount of oxygen his lungs can take in, an important measurement for an endurance athlete is extremely high some of the highest ever recorded. His heart can pump nine litres of blood per minute (at rest), compared to five for the average person. During maximal exercise, this amount goes up to around 40 litres. His resting heart rate is only 32!

Even though his heart and lungs are something extremely rare he also produces half as much lactic acid and expel it faster, allowing him to ride harder up the Tour's notoriously brutal hills and then recover quickly. With that an average person turns 20 per cent of the oxygen he breathes into muscle power. Armstrong gets 23 per cent --

www.scientificamerican.com

6.1 Using your knowledge of respiratory physiology, explain what V02 Max means.

The maximum amount of oxygen your body can use to produce maximum energy.

_____(3)

6.2 List two structures in the respiratory system that will contribute **significantly** influence gaseous exchange and therefore VO2 Max.

Capillaries and Alveoli

_____(2)

6.3 Using your understanding of **energy systems**, explain why V02 Max has a direct effect on endurance performance.

More 02 that enters the mitochondria = more ATP produced per muscle Contraction. ✓✓

The more ATP produced per muscle contraction = more force production.✓

Increased force production assists with increase performance.✓

_____

_____

_____

_____(4)

6.4 According to Source C, what is Lance Armstrong's maximum heart rate?

Over 200 bpm

_____(2)

6.5 According to Source C, what is Lance Armstrong's Cardiac Output at rest?

9 litres/min

_____

_____(2)

6.6 Calculate Lance Armstrong's **Stroke Volume** when cycling at maximum intensity. Show working.

CO= SV X HR

40 litres= SV x 200

SV= 40 Litres/ 200 bpm

SV= 40 L/ 200bpm

SV= 0.2L / beat or 200ml per beat

_____

_____(4)

6.7 List three physiological adaptations that the **respiratory system** would undergo as a result of regular cycling training for 12 months.

Capillarisation

Increased alveoli number

Respiratory muscle hypertrophy (diaphragm and intercostals)

Increase V02 max

Increase lung capacity

_____

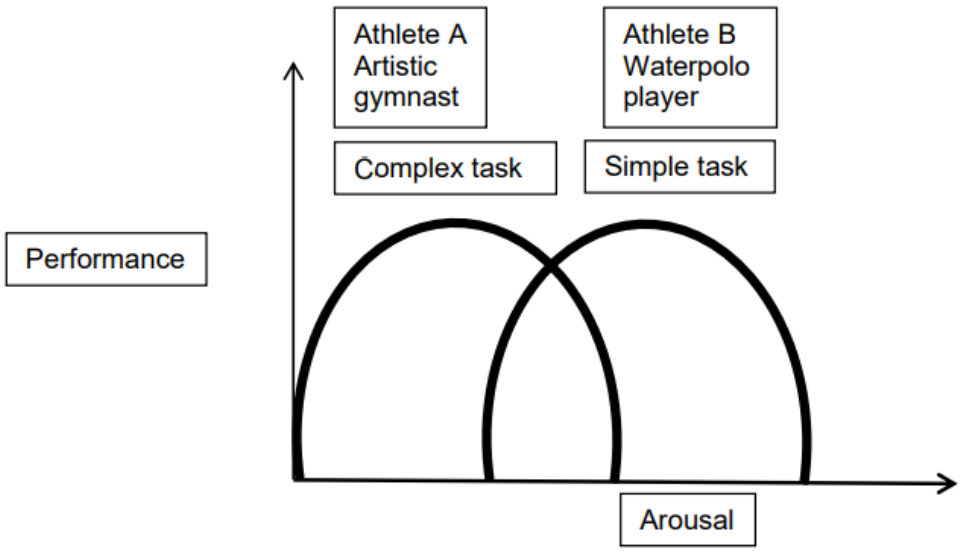
_____(3)

[20]

QUESTION 7

Examine Graph B and answer the questions that follow.

Graph B



https://www.sportpsychology.com/

7.1 Which task on Graph B requires a higher level of arousal?

Simple Task___(1)

7.2 Provide justification for your answer in 7.1

Simple tasks do not require concentration like complex tasks.
Increase arousal help with strength and speed activites.

_____

_____(3)

7.3 Explain why the two athletes depicted in Graph B might psych themselves

up very differently from each other before a major competition.

Simple tasks will aim to increase arousal because they will use a high amount of force and muscle contraction.

Complex tasks will aim to decrease arousal as complex tasks require concentration and silence to succeed.

_____(4)

7.4 Explain the difference in motor skill required to complete a complex task vs a simple task.

Complex tasks require fine motor skill (darts, pool, snooker) as accuracy contributes primarily to success.

Simple tasks require gross motor skills as high force production contributes primarily to the success of the activity.

_____

_____

_____(4)

7.5 Select two techniques and provide an explanation of how an athlete may decrease their arousal levels, if feeling overwhelmed before a big match or competition.

Deep breathing technique
Visualisation
Progressive muscle relaxation
Positive Self Talk

_____

_____

_____(4)

7.6 Provide a name for the theory exhibited in Graph B

Inverted U Hypothesis

_____(2)

[18]

QUESTION 8

ESSAY

Carefully read and examine the provided sources and the rubric. Use the sources to answer the following question.

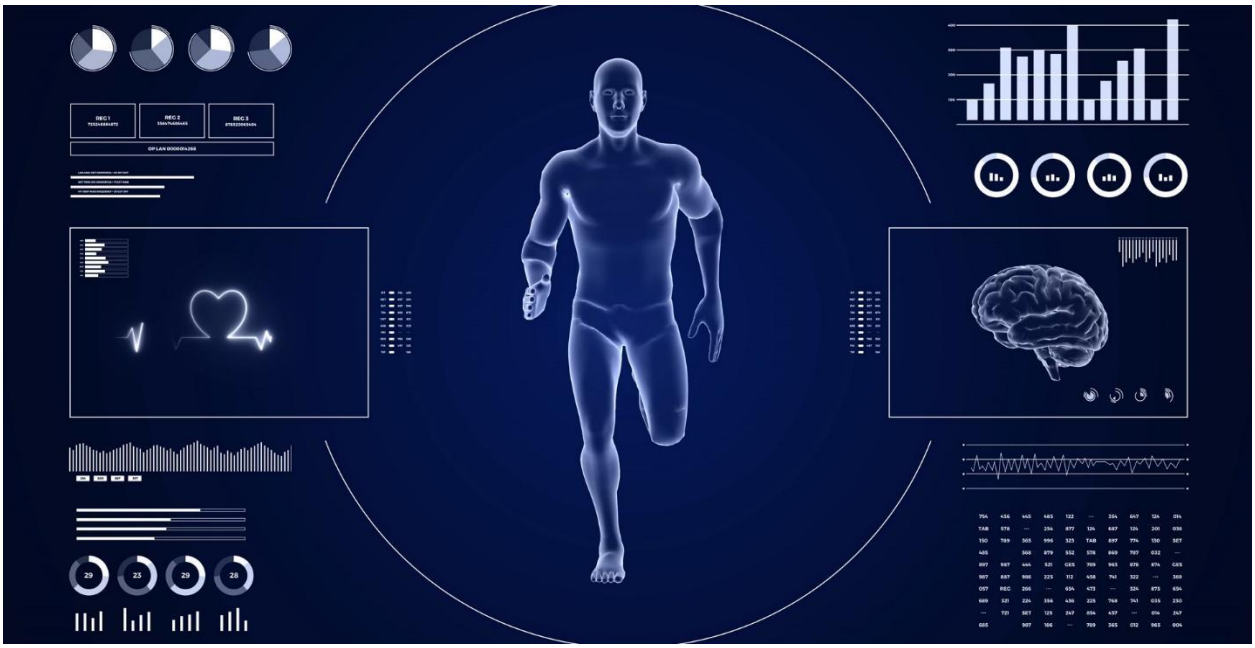
Write an essay in which you discuss/argue/debate the following:

Do you think human athletic performance has reached its limit? Provide arguments supporting and challenging this notion.

To answer this question, you are expected to:

- Examine the source material carefully and use the information in the sources to best develop your essay.
- Provide perspectives for and against the notion stated.
- Integrate your own relevant sport science knowledge into your essay.
- Use real-life examples if applicable.

(20)



https://www.mediclinicinfohub.co.za/

TOTAL

150

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.